

Coffee Brewing Outlier

08/11/2025

1 Introduction

Brewing espresso is well-known to be an extreme precise process to obtain a nice cup of drink. The recipe of an espresso, usually requires tedious dial in process. Many responsible coffee shops often need to dial in espresso shot every day because many factors such as temperature, age of coffee bean, and humidity will impact the ideal recipe to get the best cup to start the day. In this project, we are looking at the problem from a data mining perspective of view. You are given 13 espresso brewing logs recorded from DE1¹. Among these records, there exists **one espresso brewing record** that is coming from a old bag of coffee that is not for sale. The rest of the records are obtained from same bag of coffee that coffee shop owner plan to dial in and sold to customer. Unfortunately, the owner has lost the label, so we are no longer know which cup of coffee is the outlier cup. Your responsibility is to **locate the unwanted coffee cup**. Each profile file contains a set of information recorded over time, including espresso elapsed, pressure, espresso flow, espresso temperature, espresso temperature mix, espresso water dispensed, et al. Based on all these given information, you are responsible to develop a model to identify the **bean that should not appear in the dial in testing**.

The total 13 espresso logs can be accessed from link ². Feel free to search the internet to learn more about the coffee. Here are some simple links that may or (may not) be related to this problem.

Basic Espresso

Flow Profile

Espresso and Coffee Bean Age

Age and Recipe Change

and more....

In your final report, please include which cup of espresso (1 to 13), your model output as the unusual one. Some useful articles including: [2, 1, 3]

¹<https://decentespresso.com/overview>

²<https://drive.google.com/file/d/1QuvxViN5ImDmmZWY01P0f4HbDFqWCKkp/view?usp=sharing>

References

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- [3] L. Liu and Y. Qin. Detecting out-of-distribution through the lens of neural collapse. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 15424–15433, 2025.